DISTINCT F-EMBEDDINGS ARE LINEARLY INDEPENDENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The embedding form of character independence used throughout the normality and Galois-counting sections.

Editorial note. The label lemma-independence-embeddings names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter lemma-independence-characters (source lines 1675-1713, monoid version): pairwise distinct monoid homomorphisms into a field are linearly independent over the target
- Every nonzero element of the field K is a unit, and a morphism of extensions sends 0 to 0, so two F-morphisms of K that agree on the units agree on all of K

Source reference. Original reference 0CKL: sources/stacks-fields/pinned-original-source.tex, source lines 1676–1676. Prerequisite pointer invoked at 1741; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let K/F and L/F be field extensions. Let $\sigma_1, \ldots, \sigma_n : K \to L$ be pairwise distinct morphisms of F-extensions. Then $\sigma_1, \ldots, \sigma_n$ are L-linearly independent: if $\lambda_1, \ldots, \lambda_n \in L$ not all zero, then $\sum \lambda_i \sigma_i(\alpha) \neq 0$ for some $\alpha \in K$.*

Proof. Apply Lemma 0CKL to the restrictions of σ_i to the groups of units. □

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.